We should allow separate parts only when the answer to all these nine questions is yes. In some cases, we may have to keep a part separate for the reasons of market availability or modularity. In such cases, we should mark the part as a non-essential part.

Once we start questioning in this manner, we will find many avenues for reducing parts. For instance, in our case, we decided to use separate PCBs for the relays and the control circuit. The critical analysis shows that we need not have separate PCBs during manufacturing. Instead of managing two parts, we can use one PCB during manufacturing and separate the PCBs after the components are assembled. In addition, instead of using four different transistors, we can use a single integrated transistor IC that has four transistors.

The proper selection of microcontrollers can also help reduce part count. In mechanical parts, we can use flexible components to provide relative movements between parts instead of keeping them separate. Also, choosing appropriate fits and tolerances contribute to reduction of part counts. Innovation plays a vital role in simplifying the design.

Operational complexity

Operational complexity stems from the difficulty in storage, handling, and assembly of the product. We have already discussed the complexity determination of storage and handling. Within the workstation, we get various difficulties during assembly. There is a systematic process, originally invented by Cummins, to analyse and reduce assembly complexity.

Correct fitting. Every part has some dimensional tolerances. If we use parts that have opposing tolerances, assembly may become difficult. This challenge becomes more significant if the part interfaces with many other parts.

Ideally, the main part should have only one interface with the components. When this is not the case, assembly difficulties arise. Our goal should be to minimise the number of parts and the number of interfaces.

The assembly complexity index (ACI) gives an idea of assembly difficulty. If N is the number of parts and IX is the number of interfaces of xth part, then ACI is calculated as:

$ACI = \sqrt{(N \Sigma I_X)}$

Ideally, the number of parts should be the same as the theoretical minimum number of parts N_T and the number of interfaces should be $2(N_T-1)$. We can compute our target ACI as:

$ACI_T = \sqrt{[N_T \times 2(N_T-1)]}$

We can compare how far we are from the target and work accordingly. Normally, if we are within 80% of the target value, it is a good design.

Wrong assembly. During assembly, operators may omit some parts, they may fit wrong parts, or misorient the parts. A robust manufacturing system will ensure that only the correct part is presented for assembly. It will also flag a warning or alarm if some parts are missing.

There are many ways to detect missing components. The easiest is to compare the assembled item with a standard reference. The visual inspection often detects missing components easily. Alternatively, we can verify if all the parts are picked up during the assembly.

Well-designed components can help prevent wrong assembly. Designers should try to make components symmetrical that can be assembled without bothering about correct alignment. Alternatively, designers can make parts asymmetrical so that they will only mate when aligned correctly.

Handling. Difficulty in handling components creates difficulty in assembly, and will require special attention and tools. There are three kinds of difficulty in handling.

Entanglement. Parts like spring and circlip get entangled in storage bins. During assembly, the operator must take out and loosen the parts to assemble them.

Keeping the parts in separate sleeves or making some changes in design helps to avoid entanglement. If we close the ends of the springs, they do not get entangled.

Some parts, especially those which are shaped like paper cups, tend to get into each other and require some force to separate. In most cases, such nesting can be avoided by redesigning the parts.

Grasping. Slippery parts, parts with sharp edges, fragile parts, or flexible parts are difficult to grasp. The operator must pay special attention to their assembly. It is best to have some facility to make such parts drop automatically instead of handling them.

Too small. Parts that are less than 3mm in size require the use of tweezers and magnifying glasses to handle.

Insertion. When operators have to insert a component into another for assembly, they will need to spend more time. Common problems encountered during insertions include:

Difficulty in alignment or locating. Long screws, rods, or pins, etc are difficult to insert into narrow holes. They require special attention. Consider incorporating design features like chamfers or guides for the part to fall in the correct place.

Necessity to hold down the part. While assembling multiple loose parts, or when a part is to be compressed, the operators may have to hold down the parts. Such a problem may also happen while applying glue or solder.

Holding down impedes the manoeuvrability of the tool and makes the job difficult. Explore the use of